



Fig. 2: UWE-Q12 series DC-DC converter (Credit: www.murata-ps.com)



Fig. 3: AP61100Q chip (Credit: www.eepower.com)

tor. Transformer-based converters provide isolation between the input and the output.

Non-isolated. This converter is used when change in the voltage is small. It does not give protection from high electrical voltages and noise. The converters in this group include step-down (buck), step-up (boost), and buck-boost converters.

Isolated. It has a separation between input and output terminals. It has high isolation voltage properties and can block noise and interference. It also has cleaner DC output. This group includes flyback converter and forward converter.

Common parameters/specifications

It is important that a DC-DC converter operates within the range of required DC parameters for a particular application. After understanding the different parameters of the converters, system designers can choose proper DC-DC converters for their applications. The most basic DC parameters are the input voltage range, the output voltage range,

and the maximum required output current.

DC input is an important parameter in DC-DC converters. Input voltages include 12V, 24V, 48V, 110V, 280V, and 360V. DC output is another important parameter to consider in DC-DC converters. Common output voltages are 3.3V, 5V, 12V, 15V, 24V, and 48V.

Other parameters include output power, line regulation, load regulation, minimum load, operating temperature, PCB footprint area, and topology. The most common DC-DC converter topologies are buck, boost, and buck-boost (flyback). Common features for DC-DC converters include constant current, over-voltage protection, over-current protection, and short-circuit protection.

Constant current provides constant output current, irrespective of the changes in load impedance. Over-voltage protection limits or shuts down the voltage output when over-voltage condition occurs. Over-current protection limits or shuts down the current output when over-current condition occurs. Short-circuit protection protects the converter in the event of a short circuit at load end.

Some new features in DC-DC converters

There are many new features being added in some latest DC-DC converters. Apart from the wide range of input and output voltages, new features may include noise optimisation, higher efficiency, digital interface, power management, multi-channel input and output, remote control, ensuring compliance and compatibility, safety

standards, among others. Table 1 lists different DC-DC converters released/announced recently by some manufacturers.

Noise becomes an important parameter to consider for better performance and accuracy in DC-DC converters. Optimising a DC-DC converter for noise and electromagnetic interference (EMI) would enhance the performance of the power supplies in automotive and industrial applications. Electromagnetic compatibility (EMC) standards ensure proper operation of the equipment and control of EMI, so that unwanted effects are prevented.

For example, Flex's BMR683 DC-DC converter has EMC specification according to EN55022/EN55032, CISPR 22/CISPR 32, and FCC part 15J regulations. Flex's PUB-2M series has conducted EMI tests according to EN55032, CISPR 32, and FCC Part 15J. Traco Power's TMR 4 series has EMC specifications as per EN55032, and FCC Part 15, both Class A and B.

Digital interface is becoming more popular in modern converters. Traditional way of controlling DC-DC converters using analogue techniques was simple to implement. However, over the past several years, with innovation in electronic devices—such as high speed, low power consumption, better efficiency, and inexpensive semiconductors—digital control is found much more advantageous over analogue techniques. DC-DC converters with digital interfaces are digitally controlled power sources that provide accurate and adjustable levels of voltage, current, and power. Digital control may include a processor, voltage/current programming circuits, software, current shunt, and voltage/current read-back circuits.

For example, Flex's BMR683 DC-DC converter has a PMBus digital interface that enables the user to configure many aspects of the device operation through software control. It can be used with any standard two-wire I2C or SMBus host device. Rohm's BD71850MWV DC-DC converter has digital I2C interface with clock frequency up to 1MHz. Diode's AP61100Q converter has programma-